AWS EC2 Instance Deployment and Web Server Configuration

Practical Documentation

1. Objective

To deploy a Windows EC2 instance on AWS, configure IIS as a web server, host a webpage, and access it from another computer using the instance's public address.

2. Description

Amazon EC2 (Elastic Compute Cloud) provides virtual servers that can be created and managed on AWS.

In this experiment, a Windows Server EC2 instance is launched in the AWS Mumbai region. The instance is accessed through Remote Desktop Protocol (RDP), IIS is installed as the web server, and a webpage is placed in the IIS web root directory. An HTTP inbound rule is then added to allow the webpage to be accessed from another computer.

3. Topology

The topology shows the connection between the client computer, AWS, the EC2 instance, IIS, and the hosted webpage.

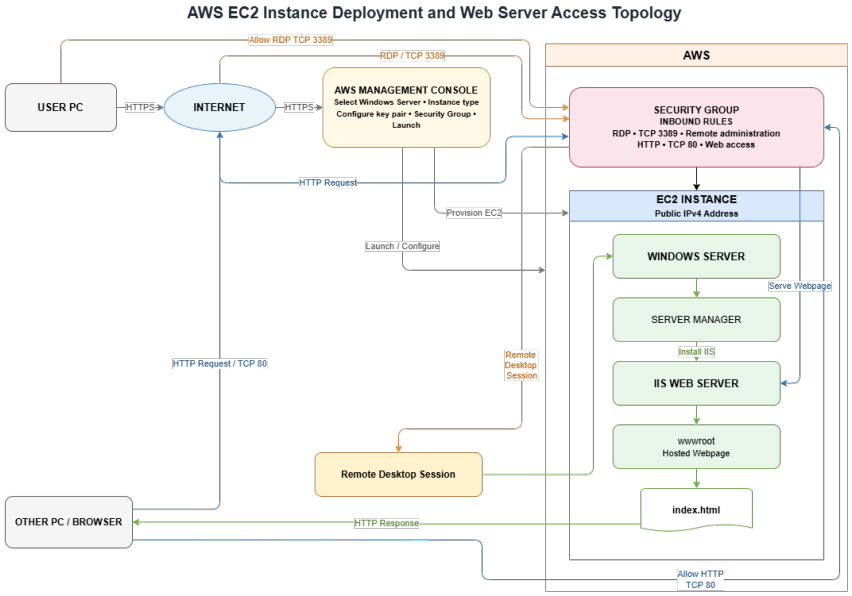


Figure 1: AWS EC2 Web Hosting Topology

4. Procedure

4.1. Step 1: Log in to the AWS Console and Open EC2

First, log in to the AWS Management Console using the AWS account credentials. From the console, open the EC2 service. The EC2 service can also be accessed from the **Compute** section of the AWS services menu.

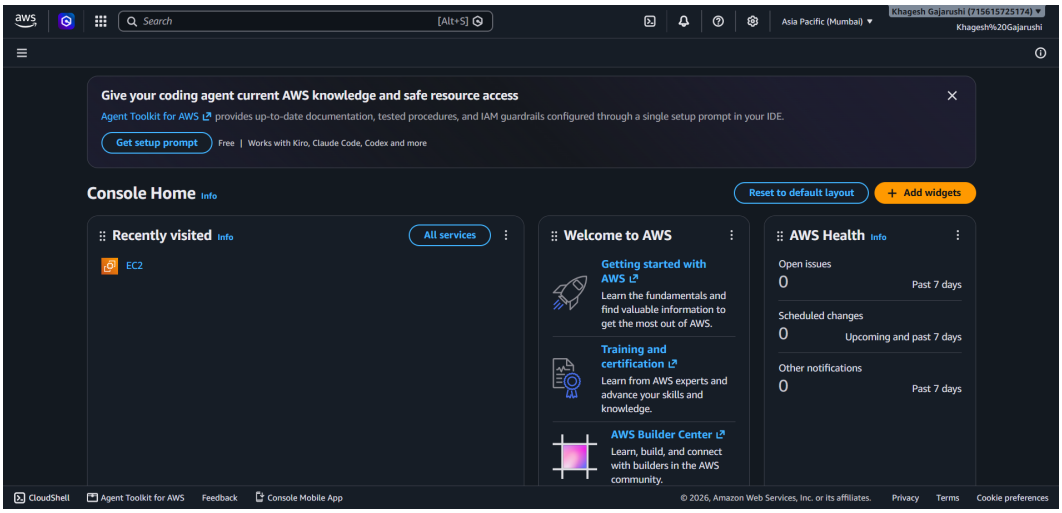


Figure 2: AWS Management Console

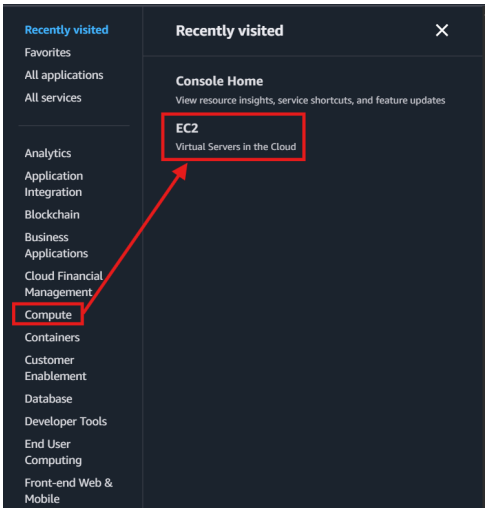


Figure 3: Opening Amazon EC2 from the AWS Services Menu

4.2. Step 2: Launch an EC2 Instance

Open the EC2 dashboard and select **Launch instance** to begin creating the virtual server.

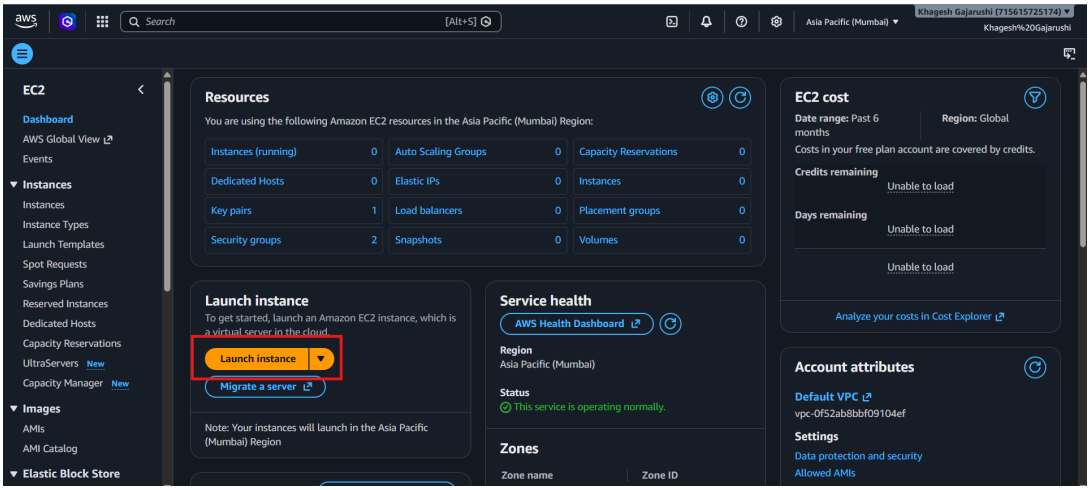


Figure 4: EC2 Dashboard and Launch Instance Option

4.3. Step 3: Configure the Instance

Enter a name for the instance and select the required operating system image. In this experiment, **Microsoft Windows Server 2022** is selected.

Next, select the required instance type. The **t3.micro** instance type is used for this experiment.

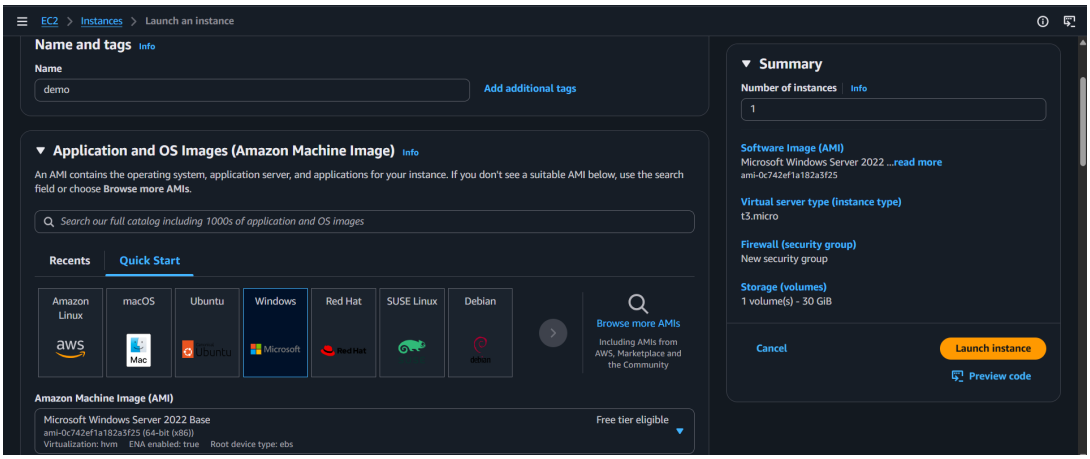


Figure 5: Instance Name and Windows Server AMI Selection

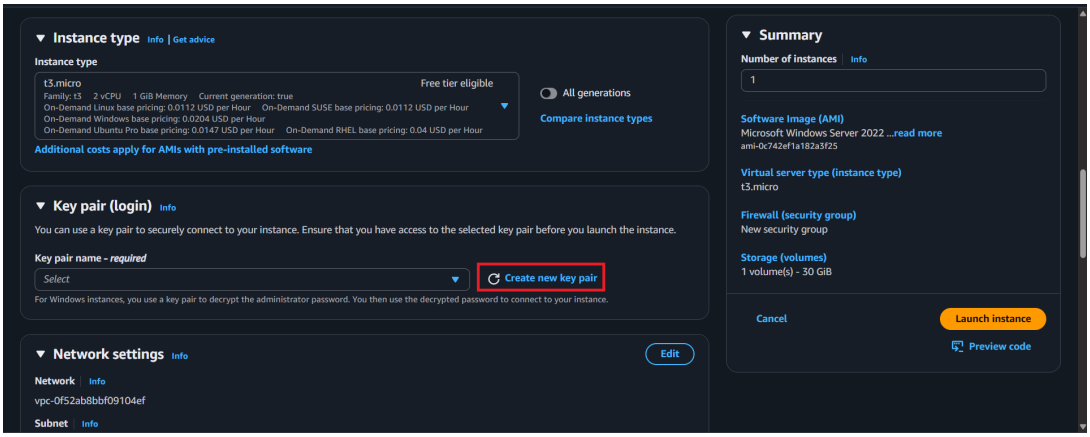


Figure 6: Instance Type and Key Pair Configuration

4.4. Step 4: Create a Key Pair

Create a new key pair for the instance. For the Windows instance, the key pair is required to retrieve the initial administrator password.

Select **RSA** as the key pair type and use the `.pem` file format. After creating the key pair, download and save the private key file securely.

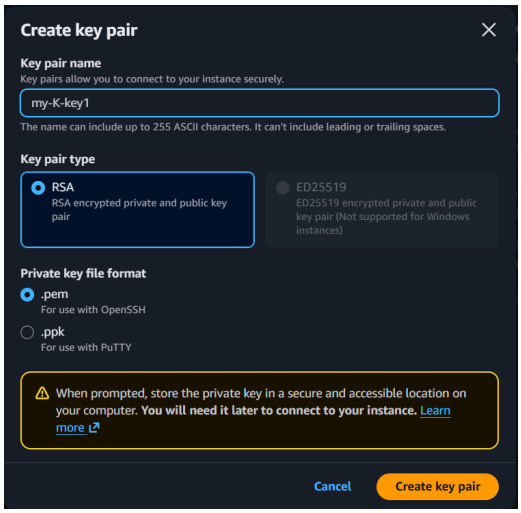


Figure 7: Creating the EC2 Key Pair

4.5. Step 5: Configure Network Settings and Launch

Review the network settings before launching the instance.

The default VPC is used, public IP assignment is enabled, and a new security group is created. The RDP rule is enabled so that the Windows instance can be accessed remotely.

HTTP access is not enabled at this stage and will be added later when the web server is configured.

After reviewing the settings, select **Launch instance**.

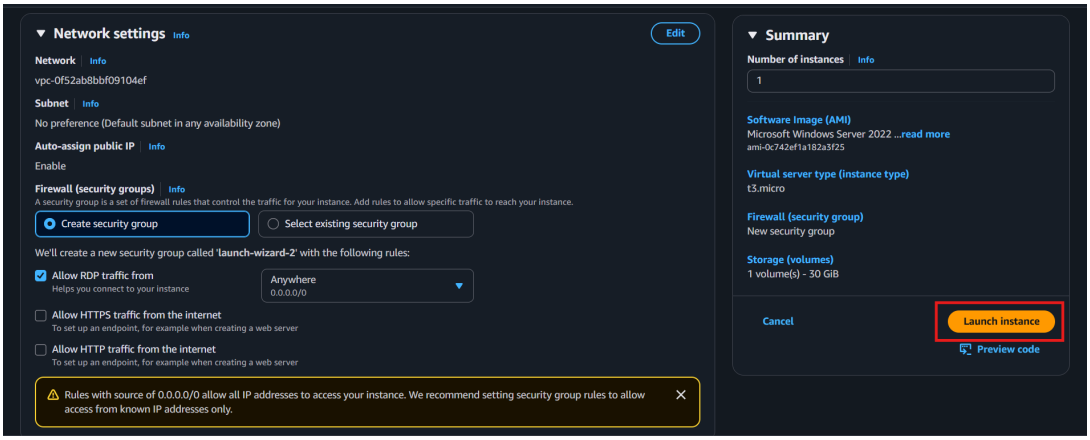


Figure 8: Network Settings and Instance Launch

4.6. Step 6: Verify the Launch

After the launch request is submitted, AWS displays a success message confirming that the EC2 instance has been created.

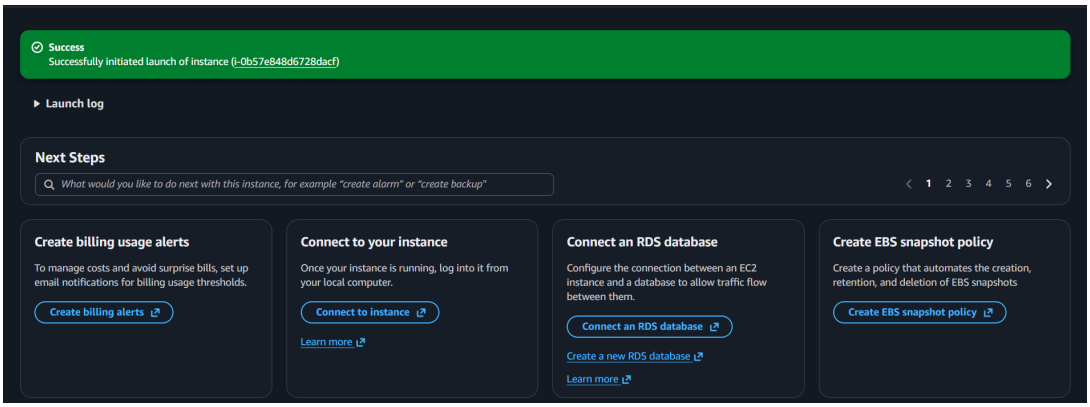


Figure 9: Successful EC2 Instance Launch

4.7. Step 7: Verify the Running Instance

Open the EC2 instance list and verify that the newly created instance is running.

The instance list shows its state, instance type, Availability Zone, and public DNS information.

Select the instance and choose **Connect**.

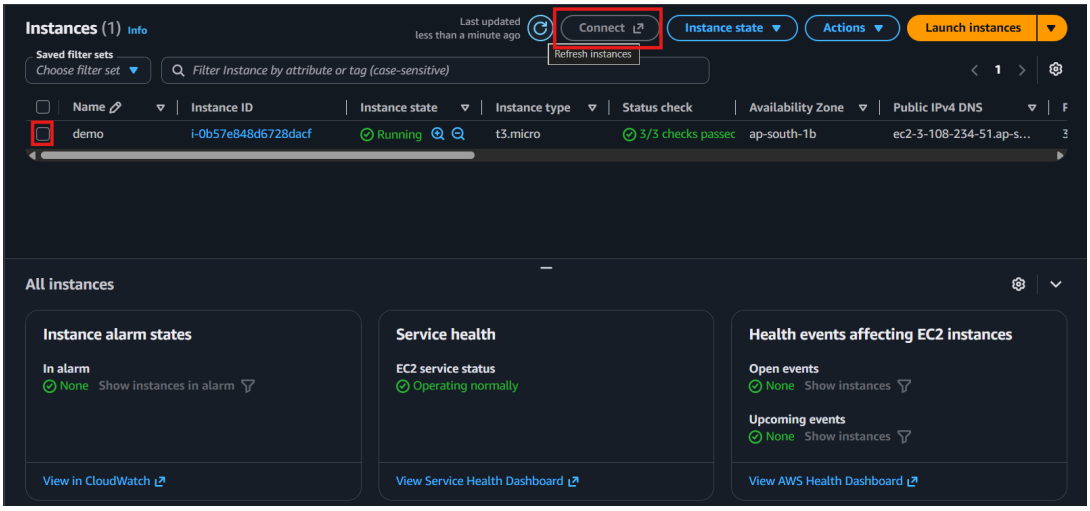


Figure 10: Running EC2 Instance

4.8. Step 8: Prepare the RDP Connection

On the Windows instance connection page, select **RDP Client**. Download the RDP shortcut file that will be used to connect to the Windows instance.

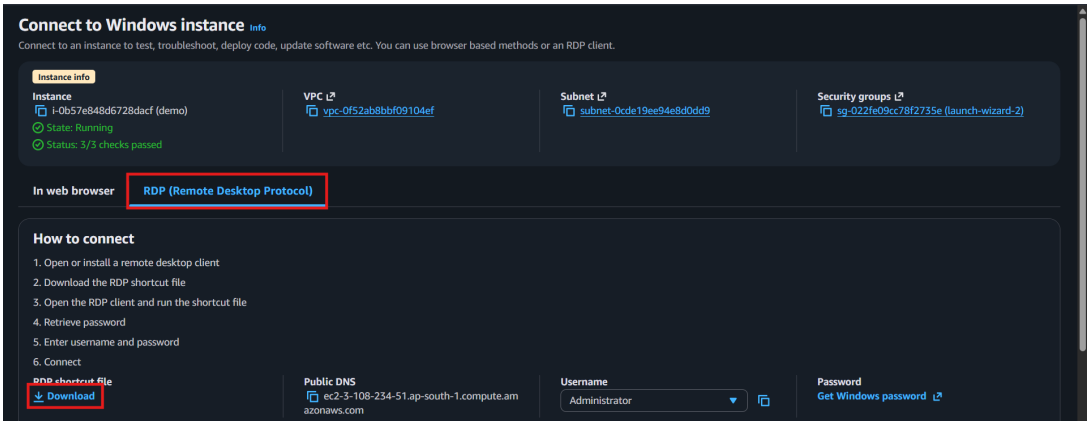


Figure 11: RDP Client Connection Details

4.9. Step 9: Open the RDP Connection

Open the downloaded RDP file. Windows may display a security warning before establishing the connection.

Review the displayed remote computer information and continue with the connection.

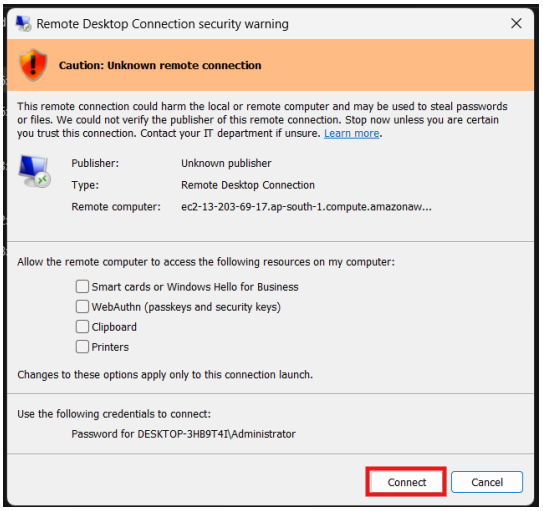


Figure 12: Remote Desktop Security Warning

4.10. Step 10: Retrieve the Windows Administrator Password and Connect

Return to the EC2 connection page and select **Get Windows password**.

AWS requires the private key created earlier to decrypt the initial Windows administrator password. Upload the downloaded `.pem` file and retrieve the password.

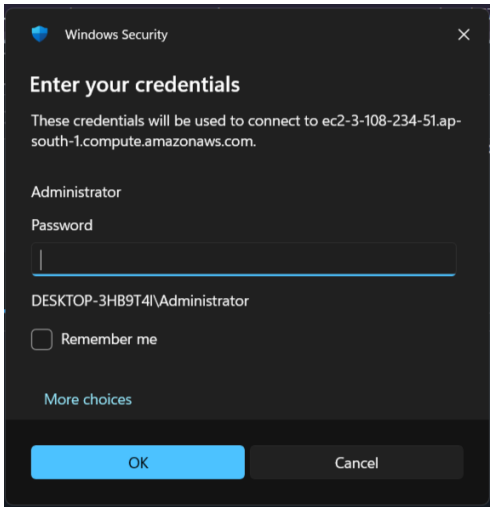


Figure 13: Get Windows Password Option

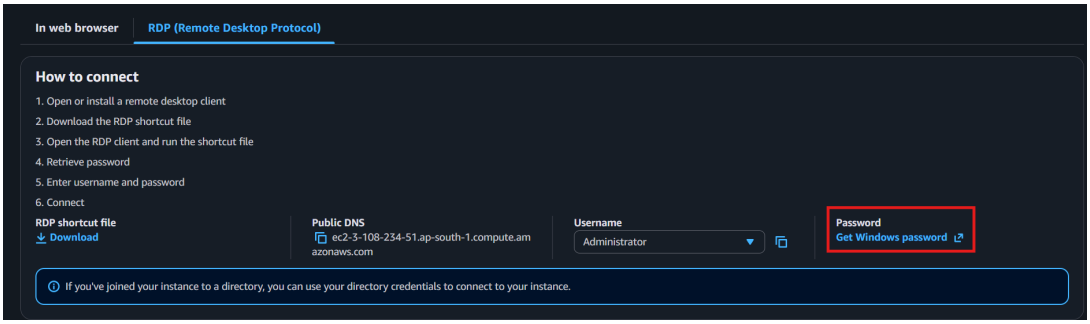


Figure 14: Uploading the Private Key File

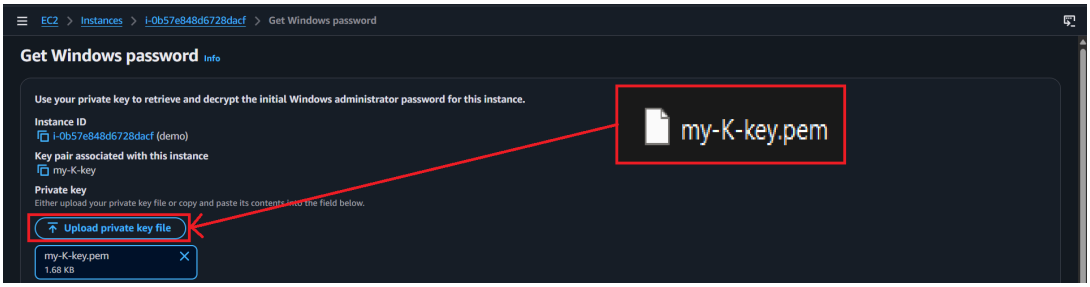


Figure 15: Retrieved Windows Administrator Password

Use the retrieved credentials to complete the RDP connection. A certificate warning may appear during the connection; continue after verifying that the connection is to the intended EC2 instance.

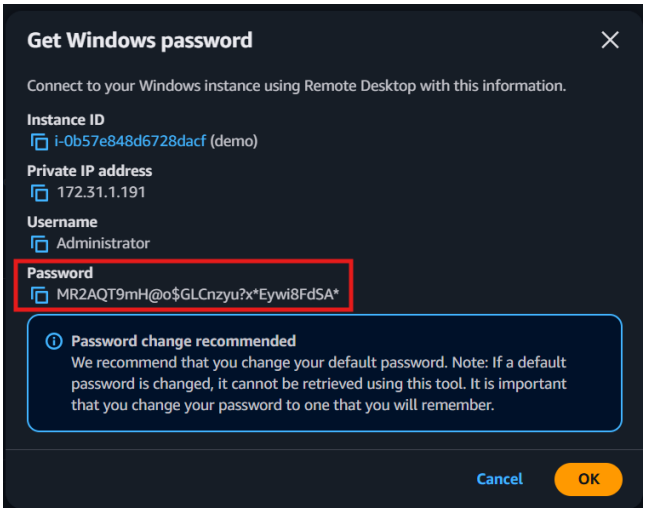


Figure 16: Remote Desktop Certificate Warning

After successful authentication, the Windows Server desktop appears.

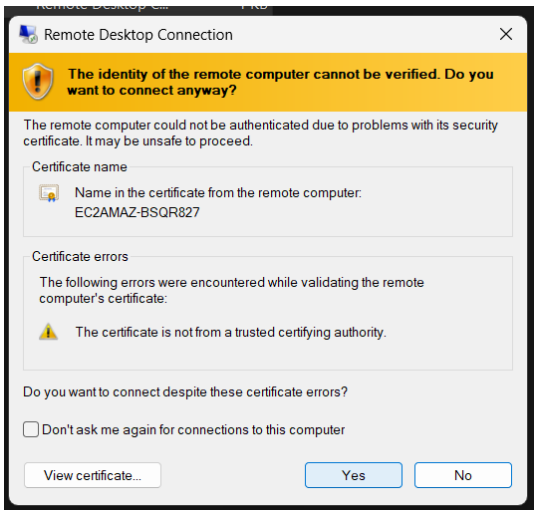


Figure 17: Connected Windows Server Instance

4.11. Step 11: Open Server Manager

Inside the Windows Server instance, open **Server Manager**. From the dashboard, select **Add roles and features** to begin installing the web server.

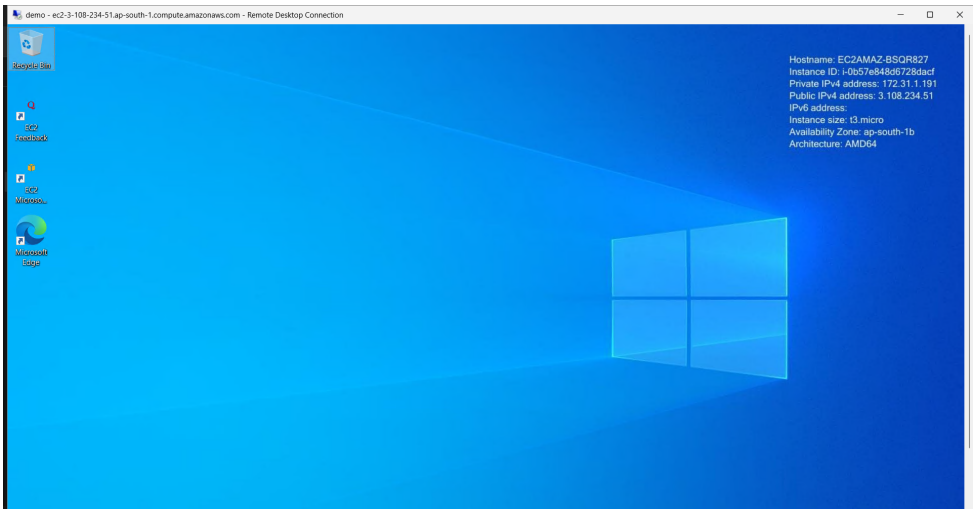


Figure 18: Opening Server Manager

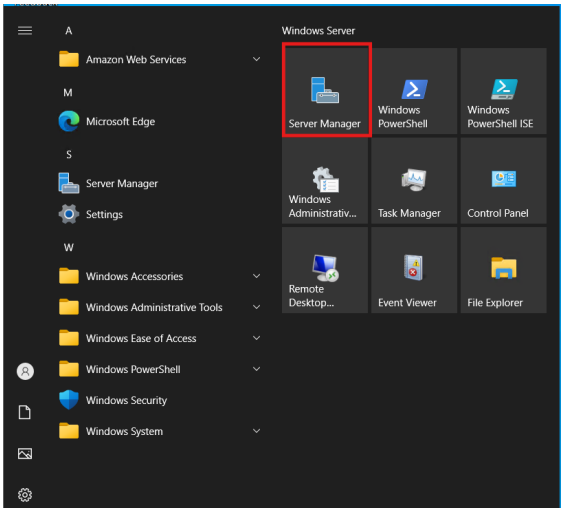


Figure 19: Server Manager Dashboard

4.12. Step 12: Install IIS

In the Add Roles and Features Wizard, proceed through the initial configuration pages until the server roles are displayed.

Select **Web Server (IIS)** as the required server role.

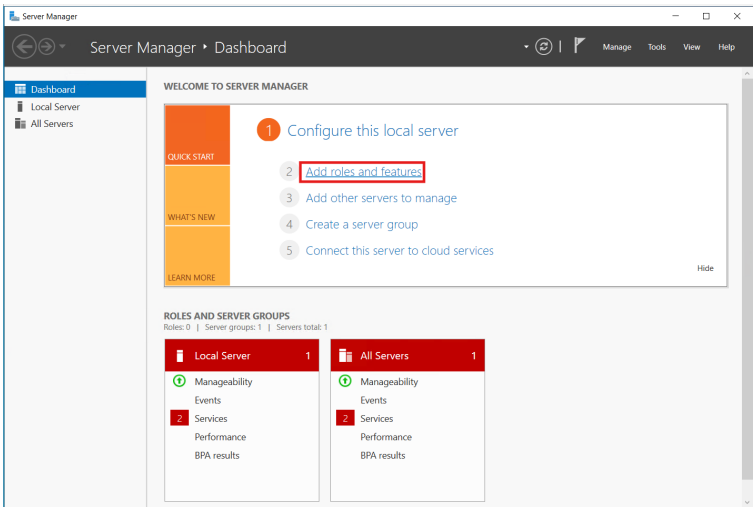


Figure 20: Selecting Web Server (IIS)

Proceed to the features section and select the required IIS-related features.

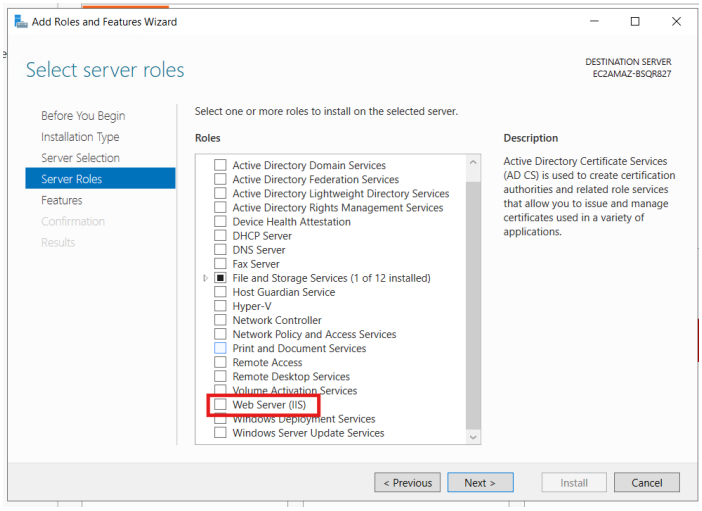


Figure 21: Selecting IIS Features

Continue to the IIS role services page and select the required IIS services.

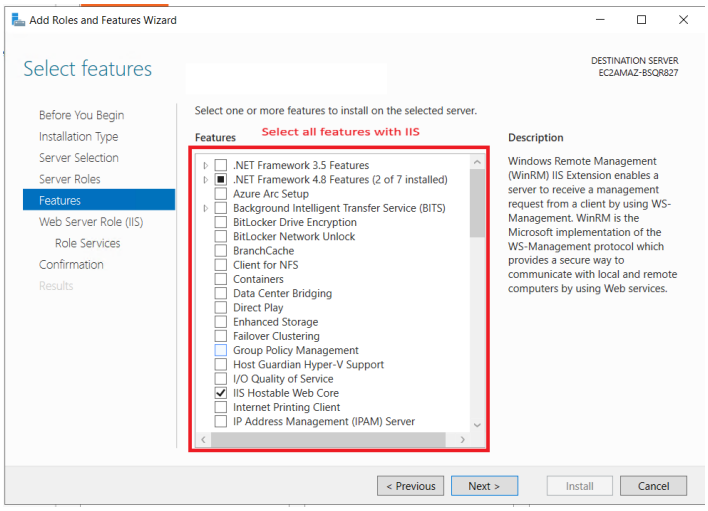


Figure 22: Selecting IIS Features

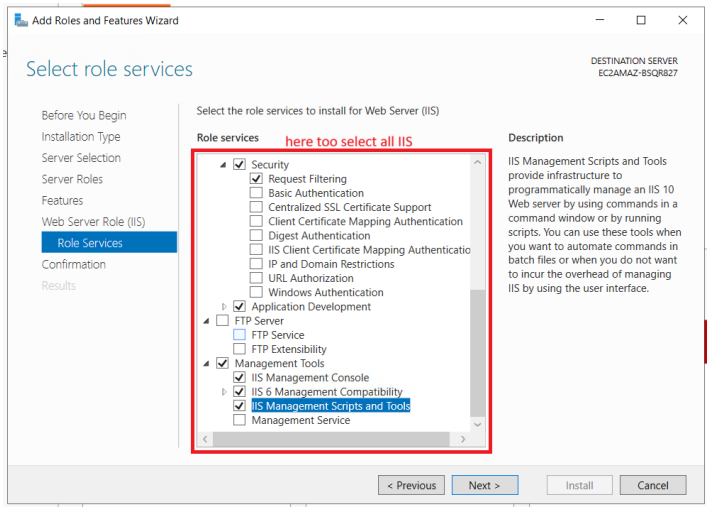


Figure 23: Selecting IIS Role Services

Proceed with the installation. Wait for the installation to complete successfully.

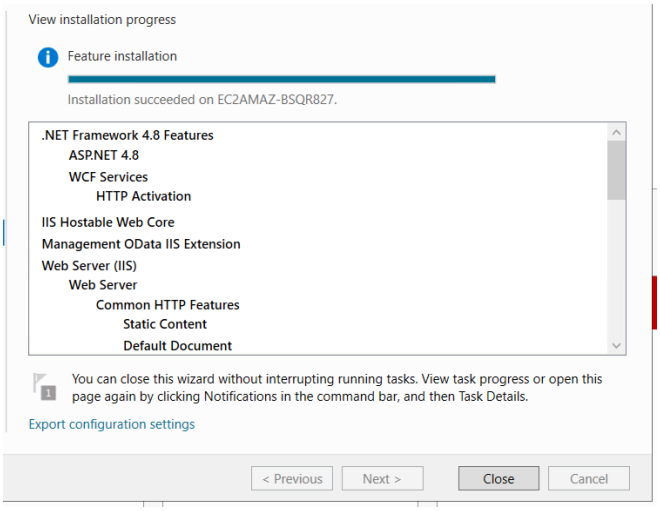


Figure 24: Successful IIS Installation

4.13. Step 13: Create and Test the Webpage

After IIS is installed, open the default web directory:

```
C:\inetpub\wwwroot
```

This is the directory from which IIS serves the webpage files.

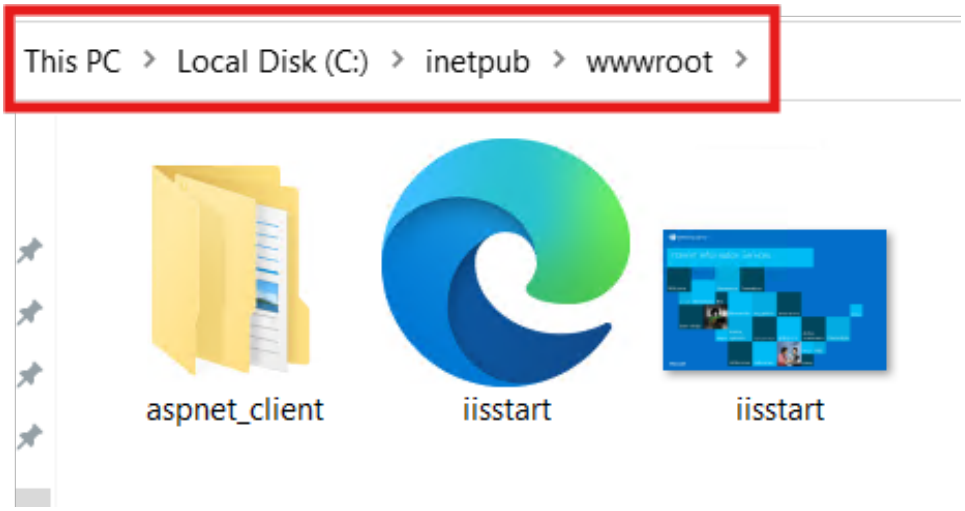


Figure 25: IIS wwwroot Directory

Open the default IIS page to confirm that the web server is working.

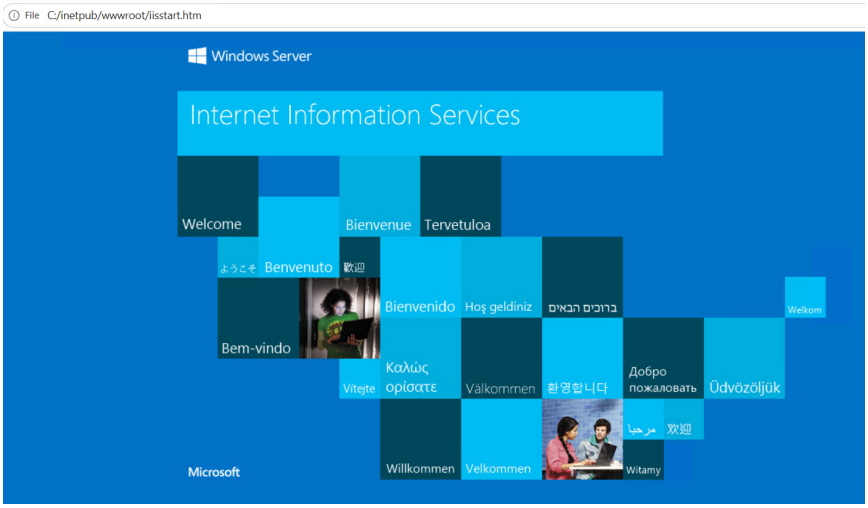


Figure 26: Default IIS Webpage

Create a new HTML file named `index.html` inside the `wwwroot` directory.

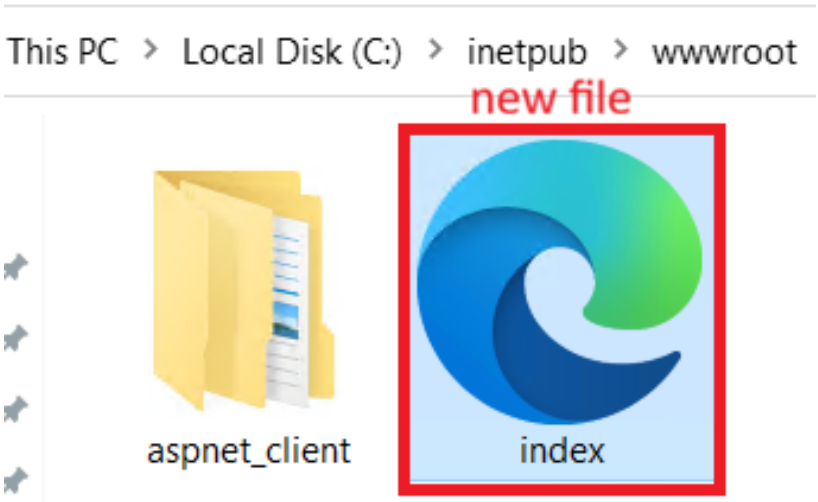


Figure 27: New index.html File

Add the webpage content and open the file through IIS to verify that the custom page is being served.

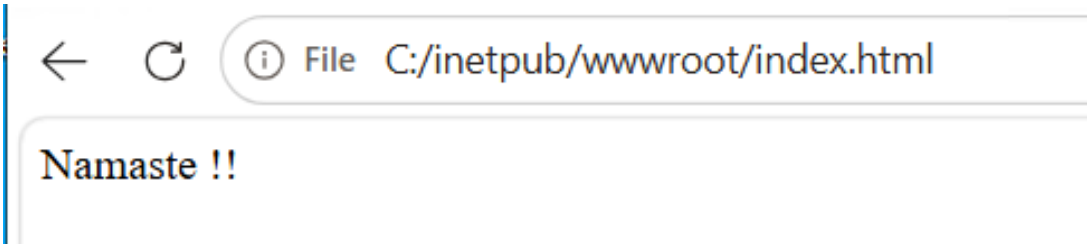


Figure 28: Custom Webpage Served by IIS

4.14. Step 14: Allow HTTP Traffic

Return to the EC2 instance in the AWS console and open its **Security** section. Select the associated security group and open its inbound rules.

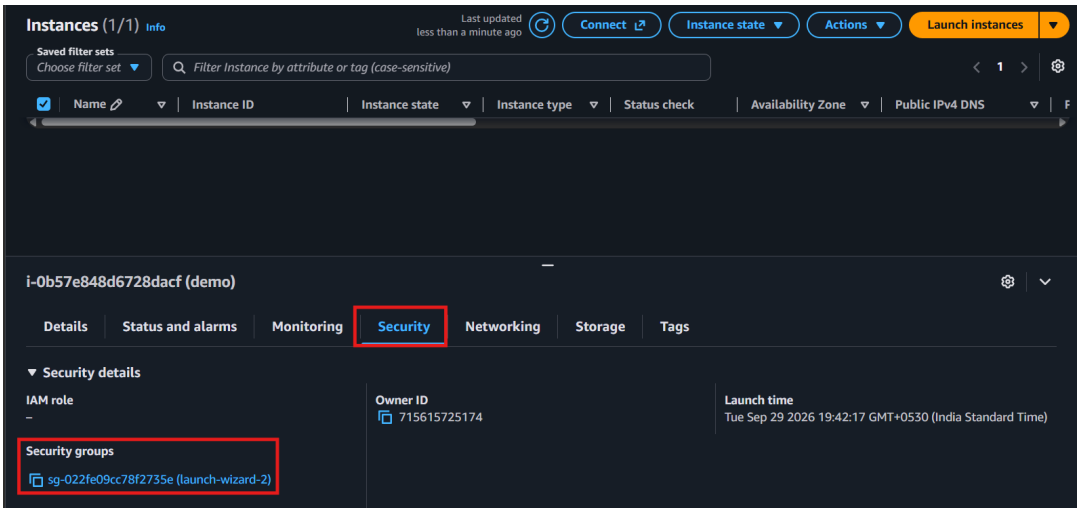


Figure 29: EC2 Instance Security Details

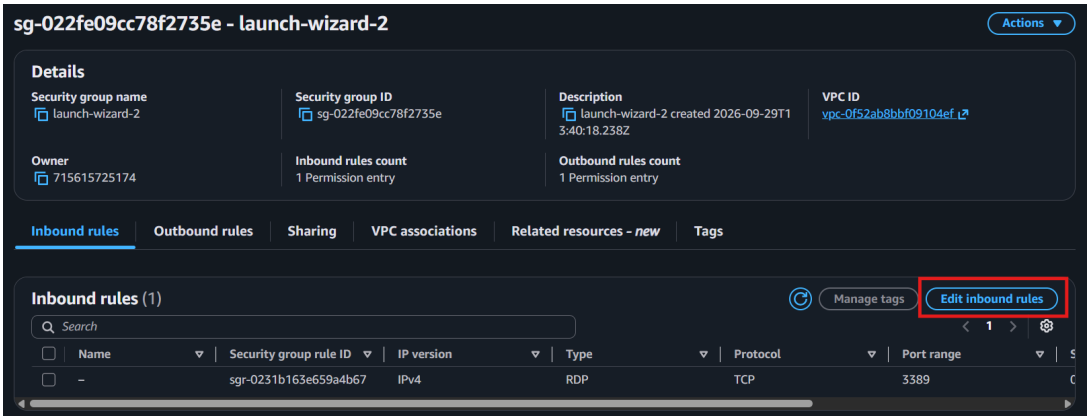


Figure 30: Security Group Inbound Rules

Select **Edit inbound rules** and add a new rule with the following settings:

- Type: **HTTP**
- Protocol: **TCP**
- Port: **80**
- Source: **0.0.0.0/0**

Save the updated rule.

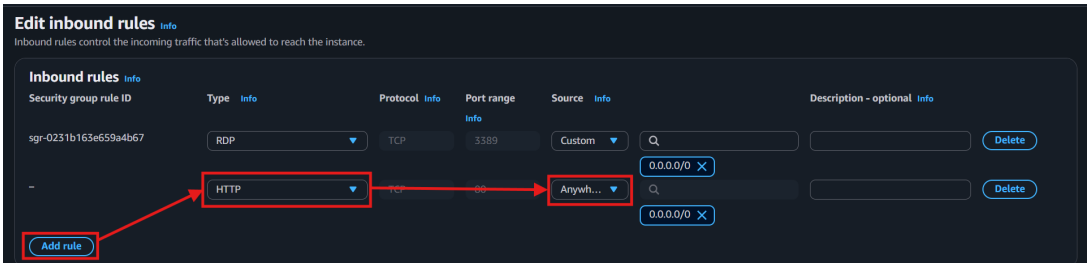


Figure 31: Adding the HTTP Inbound Rule

4.15. Step 15: Access the Webpage from Another Computer

Return to the EC2 instance details and note its public IPv4 address or public DNS name.

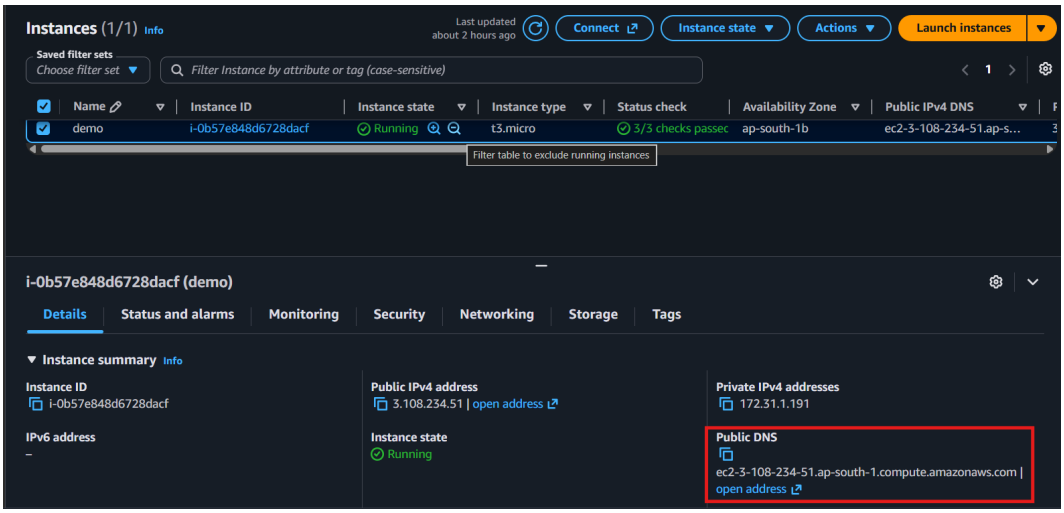


Figure 32: EC2 Public IPv4 Address and Public DNS

Open a web browser on another computer and enter the public address followed by the webpage path.

For example:

```
http://<public-address>/index.html
```

The hosted webpage should now be accessible from the other computer.

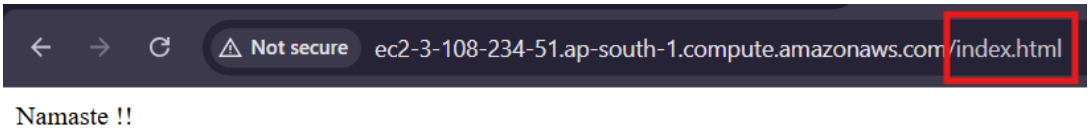


Figure 33: Webpage Accessed Using the EC2 Public Address

5. Conclusion

The Windows-based EC2 instance was successfully created on AWS and accessed using RDP. IIS was installed and configured as the web server, and a custom webpage was hosted in the `wwwroot` directory. An HTTP inbound rule was added to allow external access, and the webpage was successfully opened from another computer using the EC2 instance's public address.